

The-School-of-AI / LLM

Q

<> Code

Issues 190

Pull requests 22

Discussions

Actions

Projects

Wiki

Security

Insights

Settings

Edit

New issue

Final (PreTraining) Benchmarks #17

Open

4 / 10

#157

THE SCHOOL OF AI

theschoolofai opened last week · edited by theschoolofai

Edits Member

Team 17: Final Benchmarks & Evaluation Strategy (Pretraining Truth Gate)

Category

Evaluation / Meta

Objective (Brief)

Produce a **single, clean, defensible evaluation of the pretrained model**, while also **mapping the full benchmark landscape** required for downstream SFT, alignment, and agentic evaluation.

Team 17 owns **benchmark selection integrity** and **final pretraining evaluation execution**.

This team does **not** optimize models, tune prompts, or interpret results beyond their valid scope.

Purpose (Charter)

To ensure that:

- pretraining progress is measured **honestly**
- claims are **phase-correct**
- downstream teams inherit a **clear evaluation roadmap**
- no benchmark is misused or prematurely reported

This team prevents the most damaging failure mode in research systems:

Running the wrong benchmark at the wrong time and drawing the wrong conclusion.

Role Definition (Authoritative)

Team 17 has **two distinct, non-overlapping roles**.

ROLE A — Benchmark Cartography (Exploration Role)

Purpose

To **identify, classify, and document all relevant benchmarks** across the **entire model lifecycle**, including those **not run in this phase**.

This role informs:

- dataset choices upstream
- SFT and alignment planning (Team 18 / Team 19)
- future claims and papers

What You Must Do (Role A)

1. Survey and catalog benchmarks across categories:

- pretraining (base models)
- instruction following
- reasoning
- code
- multilingual / Indic
- agentic behavior
- safety & alignment

2. For each benchmark, document:

- what it actually measures
- which training phase it is valid for
- whether it assumes instruction tuning
- quantization tolerance
- known pitfalls / misuse risks

3. Explicitly classify benchmarks as:

- **RUN NOW** (pretraining-valid)
- **DEFERRED** (post-SFT / alignment)
- **RESEARCH-ONLY / UNSAFE**

Hard Rule (Role A)

Team 17 **does not execute** deferred benchmarks.

Discovery ≠ execution.

ROLE B — Final Pretraining Evaluation (Execution Role)

Purpose

To run **one final, clean evaluation pass** that measures **representational competence**, not behavior.

This is the **last evaluation before SFT begins**.

Benchmark Eligibility Rules (Non-Negotiable)

Benchmarks executed in this role must:

1. Be valid for **base (non-instruction) models**
2. Not require chat formatting
3. Not require alignment or preference learning
4. Be prompt-minimal or continuation-based
5. Be quantization-tolerant (INT8 / INT4)
6. Be comparable across checkpoints

Anything violating these rules is **deferred**.

Authorized Benchmark Set (Suggestive)

Team 17 is authorized to run **only the following benchmarks** for pretraining evaluation:

1) MMLU (Full, Quantized)

- Run once on the final pretrained checkpoint
- Report **category-wise results only**

- No instruction prompting
- Framed strictly as:

“Knowledge and reasoning substrate evaluation”

2) GSM8K (Continuation-Style Only)

- No chain-of-thought prompting
- No explanations
- Measures latent arithmetic structure, not reasoning skill

3) Code Completion Evaluation (HumanEval-Style, Modified)

- Completion-only
- No docstring instructions
- No natural language explanations
- Measures code structure emergence

4) Indic / Multilingual Continuation Probes

- Perplexity or continuation-based
- No translation or QA
- Presence and stability only, not fluency claims

5) (Optional) Minimal Pair / Grammar Sensitivity (BLiMP-style)

- Used as a representation health signal
- Optional, but strongly recommended if time permits

Explicitly Disallowed in This Phase

The following **must not be run or reported** by Team 17:

- Instruction-tuned MMLU variants
- MT-Bench, AlpacaEval, Arena
- Tool-use benchmarks
- Agent planning benchmarks (WebArena, SWE-bench, etc.)
- Safety / refusal benchmarks
- Preference modeling benchmarks

These are deferred to **Team 18 / Team 19**.

Responsibilities (Charter)

A) Evaluation Integrity

- No cherry-picking
- No prompt tuning to inflate scores
- No cross-phase comparisons

B) Claim Discipline

- Results must be labeled “**Pretraining-Only**”
- No behavioral or instruction-following claims
- No comparisons against instruction-tuned models

C) Upstream Feedback Loop

- Flag weak areas to:
 - data teams
 - curriculum teams
 - synthetic data teams
- Do not propose fixes; only surface signals

Inputs

- Final pretrained checkpoint(s) from **Team 12**
- Evaluation harnesses and reduced eval logic from **Team 16**
- Tokenizer artifacts from **Team 6**
- Quantization constraints

Tools

- Quantization tools (INT8 / INT4)
- Evaluation harnesses
- Reporting and plotting scripts

You Must Submit (Brief)

1. **Benchmark Landscape Report**
 - full catalog (run + deferred)
 - phase-correct classification
2. **Final Pretraining Benchmark Report**
 - raw scores
 - category-wise breakdowns
 - trend context (if available)

3. Methodology Notes

Open

Final (PreTraining) Benchmarks #17



- quantization details
- reproducibility instructions

Outputs (Charter)

- Credible, phase-correct pretraining evaluation
- Benchmark roadmap for downstream SFT/alignment
- Data-driven signals for upstream adjustments

Success Criteria

- Results are credible and defensible
- No benchmark misuse
- Downstream teams trust the evaluation
- No overclaiming in reports or papers

Failure Conditions

- Instruction-only benchmarks run on base models

- Ambiguous or inflated claims
- Results that cannot be reproduced
- Confusion between pretraining and alignment performance

Depends on (Blocking Prerequisites)

Teams 16 and 12
(and final pretrained model availability)

Blocks (Downstream Teams)

Team 18

Lifecycle

Setup / Prep

Feb 11 – Feb 12

- Finalize benchmark list
- Lock evaluation methodology

Peak

Feb 12 – Feb 16

- Run authorized benchmarks
- Produce reports

Monitor / Support

—

Done / Frozen

Feb 17

- Results frozen
- Reports archived
- Handoff complete



Sub-issues 4 of 10

	Benchmarking for Coding #61		...
	Benchmarking in Knowledge & Reasoning #62		...
	Benchmarking in Math #64		...
	Benchmarking in Multilingual/Indic #65		...
	Benchmarking in Instruction following #67		...
	Benchmarking in Agentic #68		...
	Benchmarking in Safety & Alignment #69		...
	Collating Final Benchmarking #70		...
	Benchmarking pretraining (base models) #86		...
	Benchmark for Minimal Pair / Grammar Sensitivity #102		...

Create sub-issue

- theschoolofai added this to ERA V4 Lightning LM 1.0 [last week](#)
- github-project-automation moved this to Todo in ERA V4 Lightning LM 1.0 [last week](#)
- rajankumar-spec self-assigned this [4 days ago](#)
- aiplaybookin self-assigned this [4 days ago](#)
- sardug10 self-assigned this [4 days ago](#)
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- VishakBharadwaj94 self-assigned this [4 days ago](#)
- Krishnakanth1993 self-assigned this [4 days ago](#)
- roikka self-assigned this [4 days ago](#)
- aravindchakravarti self-assigned this [4 days ago](#)
- aiplaybookin added sub-issues [3 days ago](#)
 - [Benchmarking for Coding #61](#)
 - [Benchmarking in Knowledge & Reasoning #62](#)
 - [Benchmarking in Math #64](#)
 - [Benchmarking in Multilingual/Indic #65](#)
 - [Benchmarking in Instruction following #67](#)
 - [Benchmarking in Agentic #68](#)
 - [Benchmarking in Safety & Alignment #69](#)
 - [Collating Final Benchmarking #70](#)
 - [Benchmarking pretraining \(base models\) #86](#)
- Krishnakanth1993 added a sub-issue [3 days ago](#)
 - [Benchmark for Minimal Pair / Grammar Sensitivity #102](#)
- nitin-vig linked a pull request that will close this issue [P8- MoE Architecture : Issue-141 : Add MoE cadence validation and risk decision framework #157](#) [2 days ago](#)
- github-project-automation moved this from Todo to In progress in ERA V4 Lightning LM 1.0 [2 days ago](#)



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-  narasimhatejav
-  rajankumar-spec
-  roikka
-  sardug10
-  sidrocks

Labels



No labels

Type



No type

Projects



No projects

Milestone



No milestone

Relationships



None yet

Development



Code with agent mode

 **P8- MoE Architecture : Issue-141 : Add MoE cadence validation and risk decision framework**
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